IV.a — Performance Calculations (Thermochemical)

Student worksheet (simplified) · Section IV intro + IV.a · Paraffin / liquid N_2O

The ÖTR template asks: *"İtki denklemleri ve ilgili temel parametreler ayrıntılı bir şekilde bu bölümde açıklanmalıdır. ... Seçilen yakıt-oksitleyici çiftinin termo-kimyasal hesapları bu bölümde detaylı olarak anlatılmalıdır."*

You need: Nothing: this is the first worksheet. Its results feed IV.b, IV.c and IV.d.

How to use this sheet

- Work row by row: read the **command** (left), open the **reference** (middle), then calculate and fill the blank (right).
- Values marked **Team decision**, **Team rule** or **Team estimate** are given. Find or calculate everything else yourself.
- Stop at the yellow **Advisor check** rows.
- This sheet is simplified on purpose. Your final numbers may differ slightly from the team ÖTR. That is fine.
- Do not open Hesap_Belgesi.md or the team ÖTR before you hand in. They are the answer keys.

Sources (folder Performans_Hesaplari_2026\Kaynaklar)

- [1] TEKNOFEST 2026 Şartname (project folder)
- [2] Sutton & Biblarz, Rocket Propulsion Elements, 9th ed.
- [3b] NASA RP-1311 Part II — CEA user manual
- [4b] CEA N_2O record (text file)
- [5] Karabeyoglu et al. 2005, AIAA-2005-3908
- [6] Karabeyoglu et al. 2003, AIAA-2003-1162
- [11] Huzel & Huang 1992
- n2o_sat_table.csv (folder Performans_Hesaplari_2026)
- ÖTR template: B2_-_OTR_Sablonu_2026_9O6kM.docx

Units and rules

- SI everywhere: Pa, N, kg/s, m². Only the CEA input takes bar.
- 1 bar = 10^5 Pa; 1 kcal = 4.184 kJ.
- CEA prints c*, Isp and Ivac in m/s. Isp in seconds = value in m/s ÷ g_0.
- Keep at least 5 significant figures. Round only in the report tables.
- Tool: NASA CEA 3 (pip install cea; set CEA_DATA_DIR to its data folder) or RocketCEA. Do not use CEARUN: it cannot take your own propellant entries and silently uses its own paraffin and gaseous N_2O.

Step 1 — Requirements and constants → report: Section IV (intro)

Collect the numbers the Şartname fixes. Every later step uses this page.

Do this	Reference	Calculate / write down
1.1 Open the Şartname and write down the thrust level F.	[1] Art. 5.3.3, p. 45	F = 100 N min
1.2 Write down the chamber pressure. Convert it to pascal.	[1] Art. 5.3.4, p. 45	P_c = 10 bar = 10^6 Pa (1 bar = 10^5 Pa)

Do this	Reference	Calculate / write down
1.3 Write down the vacuum specific impulse requirement. Note the wording: Isp must be **greater** than this value.	[1] Art. 5.3.5, p. 45	$I_{sp,req}$ = 220 s min
1.4 Write down the burn time.	[1] Art. 5.3.6, p. 45	t_b = 8 s
1.5 Write down standard gravity with all of its digits.	[2] Appendix 1, p. 743	g_0 = 9,81 m/s²
1.6 Use standard sea-level ambient pressure (the Şartname does not give one for B2).	**Team decision**	P_a = 101 325 Pa
1.7 Compute the total impulse for constant thrust.	[2] Section 2.1	$I_t = F \cdot t_b$ = 800 N·s min
1.8 Make the "Design requirements" table for the report: Parameter | Value | Şartname article.	ÖTR template, Section IV	Table done ☑

Watch out

- F is measured on the test stand at sea level. Later you size the throat with the **sea-level** C_F, not the vacuum C_F.
- The Isp requirement is a check you do at the end (Step 9). It is not an input.

Step 2 — Thrust equations and basic parameters → report: Section IV (intro): "İtki denklemleri ve temel parametreler"

Four definitions carry every calculation in this sheet. Write each one with its source.

Do this	Reference	Calculate / write down
2.1 Write the thrust equation. Sutton uses 1 = chamber, 2 = nozzle exit, 3 = ambient; write it with P_c, P_e, P_a. Name the two terms.	[2] p. 33, Eq. 2-13	$F = \dot{m} \cdot v_e + (P_e - P_a) \cdot A_e$ momentum term + pressure term
2.2 Write the specific impulse.	[2] p. 27–28, Eq. 2-5 and 2-6	$I_{sp} = F / (\dot{m} \cdot g_0) = c / g_0$
2.3 Write the characteristic velocity. Add one sentence: c* measures the combustion chamber only.	[2] p. 34, Eq. 2-17	$c^* = P_c \cdot A_t / \dot{m}$
2.4 Write the thrust coefficient. Add one sentence: C_F measures the nozzle only.	[2] p. 63, Eq. 3-31	$F = C_F \cdot P_c \cdot A_t$
2.5 Write the expansion ratio.	[2] p. 57, Eq. 3-19	$\varepsilon = A_e / A_t$

Do this	Reference	Calculate / write down
2.6 Combine rows 2.3 and 2.4 in two lines to show: thrust = chamber part × nozzle part.	Your own derivation	$F = \dot{m} \cdot c^* \cdot C_F$ $I_{sp} = c^* \cdot C_F / g_0$
2.7 Make a symbol list: every symbol above with its unit.	ÖTR template, "Terimler"	F [N], $\dot{m}$ [kg/s], P_c [Pa], …

Watch out

- c* is in m/s, but it is **not** a specific impulse. Never divide c* by g_0.
- P_a appears only in the pressure term of the thrust equation.

Step 3 — Propellant data for CEA → report: Section IV.a

CEA needs a chemical formula, an enthalpy and a temperature for each propellant. Paraffin and liquid N_2O are not in CEA's library in this form, so you build both entries.

Do this	Reference	Calculate / write down
3.1 Find the carbon number n of the paraffin wax (grade FR5560). Write the fuel formula.	[5] p. 12	n = 32 C_nH_{2n+2} = $C_{32}H_{64}$
3.2 Compute the heat of formation of the wax. The grain is at 298 K, so the last term of Eq. 104 is zero.	[5] p. 23, Eq. 104	$\Delta H_f = -6.713 \cdot n - 7.846$ = −222,662 kcal/mol
3.3 Convert it to kJ/mol. It stays negative.	1 kcal = 4.184 kJ	ΔH_f = −931.617,81 kJ/mol
3.4 Open the N_2O record. On line 2 read the molar mass and the heat of formation (last number, in J/mol). This value is for N_2O **gas**.	[4b] line 2	M_{N_2O} = 44,013 g/mol $\Delta H_{f,gas}$ = 82,05 kJ/mol
3.5 CEA's N_2O value (row 3.4) is for ideal-gas N_2O at 298 K. N_2O enters the motor as saturated liquid at 20 °C. Use the team value for the enthalpy change between these two states.	**Team value:** CoolProp (Lemmon & Span equation of state for N_2O)	$\Delta h_L = h_{L,sat}(293\ K) - h_{ig}(298\ K) = -11.228$ kJ/mol

Do this	Reference	Calculate / write down
3.6 Liquid N_2O enthalpy = gas value + this correction.	Rows 3.4 and 3.5	$h_{N_2O(L)} = \Delta H_{f,gas} + \Delta h_L =$ 70,822 kJ/mol
3.7 Fill in the two CEA propellant entries.	[3b] §2.3.5, p. 9	Fuel: PARAFFIN, C 32 H 64 h = -222,662 kJ/mol, T = 298.15 K Oxidizer: N2O_L, N 2 O 1, h = 72,822 kJ/mol, T = 293.15 K Both: 100 % by weight

Watch out

- Do not give CEA the gas enthalpy of N_2O. It makes T_c and c* too high.
- The heat of vaporization alone ($h_V - h_L \approx 7.5$ kJ/mol at 293 K in n2o_sat_table.csv) is not enough. It leaves out the real-gas difference between N_2O vapour at 50 bar and the ideal gas that CEA uses, and makes T_c about 18 K and c* about 7 m/s too high.
- Eq. 104 gives kcal/mol. Entering it as kJ/mol is a factor-of-four error.

Step 4 — Run CEA and choose the mixture ratio O/F → report: Section IV.a

Run CEA over a range of O/F, plot the results and pick the O/F where c is highest and flat.*

Do this	Reference	Calculate / write down
4.1 Set up a CEA "rocket" problem: equilibrium flow, chamber pressure P_c, the two propellants from Step 3, SI units.	[3b] p. 143, Example 8; P_c from Step 1	P_c = 10 bar
4.2 Run O/F = 2, 3, 4, … 12. Use area ratio ε = 3 for now (you choose ε in Step 6; c* does not depend on ε). Record a table.	CEA output: rows T (chamber), CSTAR, Ivac	O/F | T_c (K) | c* (m/s) | $I_{sp,vac} = Ivac/g_0$ (s)
4.3 Find the O/F with the highest c*. Add runs every 0.25 for ± 1 around it.	Your table	O/F at max c* = 7 c^*_{max} = 1486 m/s
4.4 Plot T_c, c* and $I_{sp,vac}$ against O/F in one figure.	Report figure	Figure saved ☑
4.5 Find the plateau: the O/F range where c* is within 0.3 % of its maximum.	**Team definition**	$c^* \geq 0.997 \cdot c^*_{max}$ → O/F from _____ to _____

Do this	Reference	Calculate / write down
4.6 Choose the design O/F inside the plateau, on its high-O/F side. Reason: the tank pressure drops during the burn, so O/F falls.	**Team rule**	O/F = 7

■ Advisor check: Show your table, figure and chosen O/F to the advisor before Step 5.

Watch out
- Choose O/F with c* (or Isp), not with T_c. T_c peaks at a higher O/F.
- O/F is a mass ratio ($\dot{m}_{ox}/\dot{m}_f$).
- Check that CEA is set to bar and m/s, not psia and ft/s.

Step 5 — Loss factors: ηc* and λ → report: Section IV (intro) and IV.a

A real motor performs worse than CEA predicts. ηc (combustion) multiplies c*; λ (conical nozzle) multiplies the momentum part of C_F.*

Do this	Reference	Calculate / write down
5.1 Read which c* efficiencies were measured in paraffin motors.	[6] p. 7	Measured range: 103 % to 100 %
5.2 Use the team value of ηc*. Write one sentence on why, and that it must be measured in the hot-fire test.	Team decision	η_{c^*} = 0.90
5.3 Read the typical half-angle of a conical nozzle. Use the team value.	[11] p. 75; [2] p. 75	α = 15° (team decision)
5.4 Compute the divergence factor of a conical nozzle.	[2] p. 75, Eq. 3-34	λ = (1 + cos α) / 2 = 0,9829

Watch out
- λ multiplies only the CEA C_F (momentum part), never the pressure term.
- η_{c^*} multiplies only c*. It never enters C_F.

Step 6 — Choose the nozzle expansion ratio ε → report: Section IV.a

For a sea-level test, a larger ε gives more vacuum Isp but less sea-level thrust, and too large an ε makes the flow separate inside the nozzle.

Do this	Reference	Calculate / write down
6.1 Find the optimum ε (where $P_e = P_a$). In CEA, add an exit pressure ratio P_c/P_a; CEA prints the matching Ae/At.	[2] p. 59 (optimum expansion)	P_c/P_a = _____ ε_{opt} = _____
6.2 Run CEA at your O/F with ε = 2.5, 3.0, 3.5, 4.0 and 4.5. From each EXIT column read P (bar), CF and Ivac. Use c* from the same run.	CEA output, EXIT columns	Table: ε | P_e | CF | Ivac

Do this	Reference	Calculate / write down
6.3 For each ε compute P_e/P_a. Below 0.4 the flow may separate.	[2] p. 68 (flows full down to about 10–40 % of P_a); team limit 0.4	P_e/P_a = ____ for each ε
6.4 For each ε compute the real sea-level thrust coefficient.	Thrust eq. (Step 2) divided by $P_c \cdot A_t$; λ from Step 5	$C_{F,sl} = \lambda \cdot CF + \varepsilon \cdot (P_e - P_a)/P_c$
6.5 For each ε compute the real vacuum specific impulse.	η_{c^*}, λ from Step 5; g_0 from Step 1	$I_{sp,vac} = \eta_{c^*} \cdot c^* \cdot (\lambda \cdot CF + \varepsilon \cdot P_e/P_c) / g_0$
6.6 Choose ε: the smallest ε where $I_{sp,vac}$ is above $I_{sp,req}$ by at least 3 % and $P_e/P_a \geq 0.4$.	[1] Art. 5.3.5; Step 1; Team rule: margin ≥ 3 %	ε = _____ margin = $(I_{sp,vac} - I_{sp,req}) / I_{sp,req}$ = _____ %
■ Advisor check: Show your ε table and your choice to the advisor before Step 7.		
Watch out • $C_{F,sl}$ must fall when ε grows beyond ε_{opt}. If it does not, you forgot P_a. • CEA's equilibrium result is the optimistic limit. If the nozzle flow is closer to frozen, $I_{sp,vac}$ is about 2 % lower. This is why the margin must be at least 3 %. • Use the same P_c, P_a and O/F for every ε. Pressures in the same unit inside each fraction.		

Step 7 — Design-point CEA output → real c* and C_F → report: Section IV.a

One final CEA run at your P_c, O/F and ε gives every thermochemical number the report needs.

Do this	Reference	Calculate / write down
7.1 Run CEA at P_c, your O/F and your ε. Save the full output: it goes into the report appendix. Then check the reactant table at the top of the output: the ENERGY column must show your two enthalpies from Step 3, × 1000. If it does not, CEA used its own library data.	Steps 1, 4, 6	Output saved ☐ Reactant energies checked ☑
7.2 From the CHAMBER column read temperature, molar mass and γ.	CEA rows T, M, GAMMAs	T_c = 3000 K M = 24.7 kg/kmol γ = 1.25
7.3 Read c* (CSTAR). From the EXIT column (Ae/At = your ε) read CF, P and Ivac. Convert P to Pa.	CEA rows CSTAR, CF, P, Ivac	c* = 1500 m/s, CF = 1.265 P_e = _____ Pa, Ivac = _____ m/s
7.4 List the five largest mole fractions in the chamber.	CEA "MOLE FRACTIONS"	_____
7.5 Check that you read the right column.	Should be equal	$CF + \varepsilon \cdot P_e/P_c$ = _____ Ivac / c* = _____

Do this	Reference	Calculate / write down
7.6 Compute the real characteristic velocity.	η_{c^*} from Step 5	$c^*_{real} = \eta_{c^*} \cdot c^* =$ 1500 ≈ m/s
7.7 Compute the real sea-level and vacuum thrust coefficients.	λ from Step 5	$C_{F,sl} = \lambda \cdot CF + \varepsilon \cdot (P_e - P_a)/P_c =$ 1,165 $C_{F,vac} = \lambda \cdot CF + \varepsilon \cdot P_e/P_c =$ 1,497

Watch out
- CEA's "CF" and "Isp" rows assume $P_e = P_a$. They are neither sea-level nor vacuum values.
- Ivac is in m/s. Divide by g_0 to get seconds.

Step 8 — Throat size, mass flow and flow split → report: Section IV.a (values feed IV.b–IV.d)

The thrust requirement fixes the throat; the throat and c fix the propellant flow.*

Do this	Reference	Calculate / write down
8.1 Compute the throat area with the sea-level C_F.	[2] p. 63, Eq. 3-31	$A_t = F / (C_{F,sl} \cdot P_c) =$ 6,2·10⁻⁵ m² = 62 mm²
8.2 Compute the throat diameter.	Area of a circle	$d_t = \sqrt{(4 \cdot A_t/\pi)} =$ 15 mm 10,2
8.3 Compute the total mass flow with the real c*.	[2] p. 34, Eq. 2-17	$\dot{m} = P_c \cdot A_t / c^*_{real} =$ 0,6467 g/s
8.4 Split the flow into oxidizer and fuel.	[2] p. 194, Eq. 6-3 and 6-4	$\dot{m}_{ox} = \dot{m} \cdot (O/F)/(1 + O/F) =$ 0,47 g/s $\dot{m}_f = \dot{m}/(1 + O/F) =$ 0,05858 g/s
8.5 Check: rebuild the thrust from your numbers.	Step 2: $F = \dot{m} \cdot c^* \cdot C_F$	$\dot{m} \cdot c^*_{real} \cdot C_{F,sl} =$ 100 N (should equal F)

Watch out
- P_c must be in Pa. With bar, A_t comes out 100 000 times too large.
- Do not round A_t. Round d_t only for the drawing.

Step 9 — Specific impulse and the Şartname check → report: Section IV.a

Does the real motor meet the vacuum Isp requirement, and how much combustion loss can it take?

Do this	Reference	Calculate / write down
9.1 Compute the sea-level specific impulse.	[2] p. 27, Eq. 2-5	$I_{sp,sl} = F / (\dot{m} \cdot g_0) =$ 256 s
9.2 Compute the vacuum specific impulse.	Step 2: $I_{sp} = c^* \cdot C_F/g_0$	$I_{sp,vac} = c^*_{real} \cdot C_{F,vac} / g_0 =$ 256 s

Do this	Reference	Calculate / write down
9.3 Check the requirement and write the margin.	[1] Art. 5.3.5	$I_{sp,vac} > I_{sp,req}$? ☑ yes ☐ no margin = 14,06 %
9.4 Find the lowest ηc* that still passes. Use the CEA c* here, not the real one.	Row 9.2 solved for η_{c^*}	$\eta_{c^*,min} = I_{sp,req} \cdot g_0 / (c^* \cdot C_{F,vac})$ = 0,90
9.5 Make the results table for the report.	Steps 4–9	O/F, ε, c* (CEA / real), $C_{F,sl}$, $C_{F,vac}$, A_t, d_t, $\dot{m}$, $\dot{m}_{ox}$, $\dot{m}_f$, $I_{sp,sl}$, $I_{sp,vac}$, $\eta_{c^*,min}$
■ Advisor check: Show the results table to the advisor before writing the report.		
Watch out • $I_{sp,sl}$ is always smaller than $I_{sp,vac}$. Never quote the vacuum value for the sea-level test. • Use g_0, not the local gravity of the test site.		

Step 10 — Write it in the report → report: Section IV (intro) + IV.a

Write the text only after the last advisor check. Follow this order.

Write this	Template requirement	Use results from
10.1 Write one intro paragraph: what Section IV covers.	ÖTR template, Section IV	—
10.2 Add the design requirements table.	Template, Section IV	Step 1
10.3 Under "İtki denklemleri ve temel parametreler", write the numbered equations with sources, then the loss factors η_{c^*} and λ and why you chose them.	Template: "İtki denklemleri … ayrıntılı bir şekilde açıklanmalıdır"	Steps 2 and 5
10.4 Write how CEA was used (equilibrium, tool) and how you got the propellant data (formula, enthalpies).	Template IV.a: "termo-kimyasal hesapları … detaylı olarak anlatılmalıdır"	Step 3
10.5 Add the O/F table (8–10 rows), the figure and a paragraph on why you chose your O/F.	Template IV.a	Step 4
10.6 Add the design-point table: T_c, M, γ, c*, P_c/P_e, $I_{sp,vac}$ (CEA), main species.	Template IV.a	Step 7
10.7 Add the ε table and a paragraph on your choice.	Template IV.a	Step 6

Write this	Template requirement	Use results from
10.8 Add the real performance table and one sentence on $\eta_{c^*,min}$ and how the hot-fire test will measure c* ($c^* = P_c \cdot A_t / \dot{m}$).	Template IV.a	Steps 8 and 9
10.9 Put the CEA input and full output in the appendix.	Template, appendices	Step 7

Watch out

- Number every equation and cite every source as [n]. Every symbol goes into "Terimler" with its unit.
- The report is in Turkish: use the template headings and decimal commas.

Hand in → report: all

Tick each item before you hand in.

Item	From	Done
1 Calculation file (Excel or similar): every blank of Steps 1–9, with formula and unit.	Steps 1–9	☐
2 CEA input and full design-point output.	Step 7	☐
3 O/F table and figure; ε table.	Steps 4 and 6	☐
4 Results table (its $\dot{m}_{ox}$, $\dot{m}_f$, A_t are the inputs of IV.b, IV.c and IV.d).	Step 9	☐
5 Report text for Section IV intro and IV.a, in the template.	Step 10	☐
6 Three advisor checks done (O/F, ε, results).	Steps 4, 6, 9	☐